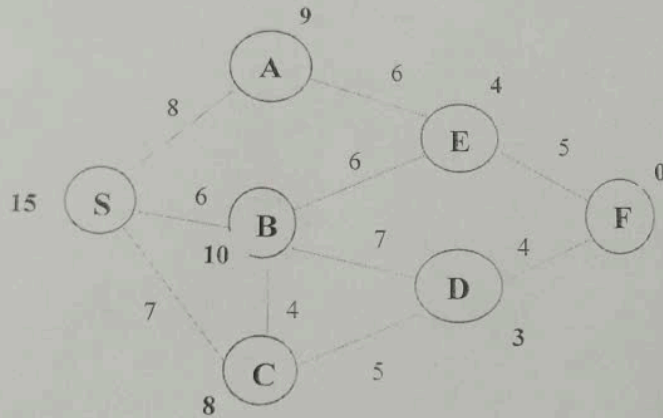


Final exam (2 pages)

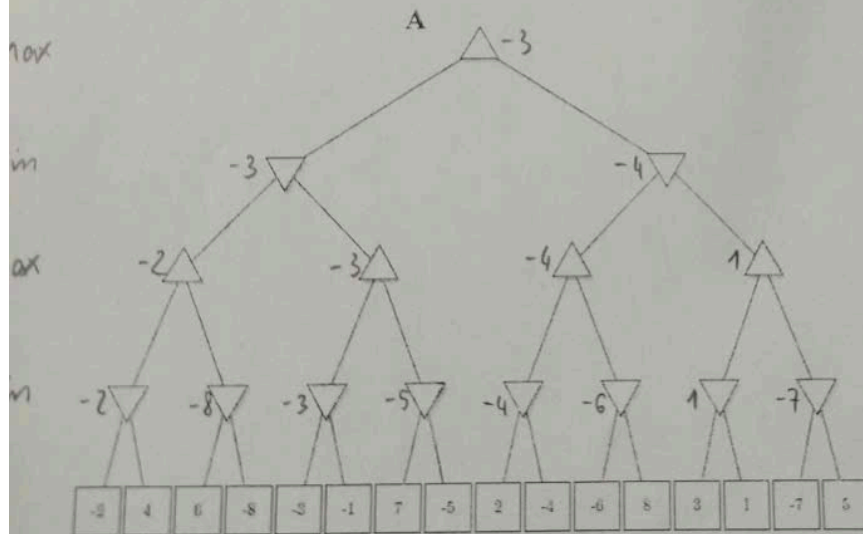
Class INT3401E (Time: 90 minutes)

(Allow using paper documents)

Question 1. Given the graph below. The length of the edge is the corresponding value on the edge. These heuristics function for the given graph by annotating each node with a value respectively. Find the shortest path from S to F by A* algorithm?



Question 2: Given game tree below (A root is Max):



1) Calculate the value of A root node by Min-Max algorithm?

2) Calculate value of A root node by α - β pruning algorithm? Leaf nodes for pruning (describes alpha, beta values)?

3) Can the nodes be ordered in such a way that alpha-beta pruning can cut off more branches? If so, give the order.

Question 3: We want to build a naive bayes sentiment classifier (two classes: **Not Politics** and **Politics**) using add-1 laplace smoothing ($k=1$). Here is our training set:

Statement	Category
basketball is a great game to play	Not Politics
the election is over	Politics
very clean debate	Politics
a close but forgettable race	Not Politics
the election is a race	Politics

Test Set:

a very close election: ? (Not Politics or Politics)

Question 4: Using a Neural Network feed-forward full connection with 1 hidden layer (including 10 neurons) to solve the problem of recognizing digits (0→9) in images. Suppose the input is an image of size 8x8

- Design the neural network above?
- Write an efficient back-propagation algorithm for the above neural network.

Question 5: Analyzing the aspects of AI Ethics for the ChatGPT system?

p/hiệu các khía cạnh của đạo đức AI cho hệ thống ChatGPT